

LSPR23: A novel IDS dataset from the largest live-fire cybersecurity exercise

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ARTICLE INFO

Dataset link: [10.5281/zenodo.8042347](https://doi.org/10.5281/zenodo.8042347)

MSC:

68T01

68U01

68U35

Keywords:

Cybersecurity components

Intrusion detection

IDS dataset

Artificial intelligence

Autonomous agents

ABSTRACT

Cybersecurity threats are constantly evolving and becoming increasingly sophisticated, automated, adaptive, and intelligent. This makes it difficult for organizations to defend their digital assets. Industry professionals are looking for solutions to improve the efficiency and effectiveness of cybersecurity operations, adopting different strategies. In cybersecurity, the importance of developing new intrusion detection systems (IDSs) to address these threats has emerged. Most of these systems today are based on machine learning. But these systems need high-quality data to “learn” the characteristics of malicious traffic. Such datasets are difficult to obtain and therefore rarely available.

This paper advances the state of the art and presents a new high-quality IDS dataset. The dataset originates from Locked Shields, one of the world’s most extensive live-fire cyber defense exercises. This ensures that (i) it contains realistic behavior of attackers and defenders; (ii) it contains sophisticated attacks; and (iii) it contains labels, as the actions of the attackers are well-documented.

The dataset includes approximately 16 million network flows, [F3] of which approximately 1.6 million were labeled malicious. What is unique about this dataset is the use of a new labeling technique that increases the accuracy level of data labeling.

We evaluate the robustness of our dataset using both quantitative and qualitative methodologies. We begin with a quantitative examination of the Suricata IDS alerts based on signatures and anomalies. Subsequently, we assess the reproducibility of machine learning experiments conducted by Känzig et al., who used a private Locked Shields dataset. We also apply the quality criteria outlined by the evaluation framework proposed by Gharib et al.

Using our dataset with an existing classifier, we demonstrate comparable results (F1 score of 0.997) to the original paper where the classifier was evaluated on a private dataset (F1 score of 0.984)

1. Introduction

With the increasing frequency and complexity of cyber attacks, the need for robust and reliable cybersecurity measures has intensified, prompting organizations to explore innovative solutions, including integrating artificial intelligence (AI) for a more effective response to cyber attacks. Research and development efforts aim to create tools for threat detection, attack pattern analysis, and the implementation of real-time defensive actions [1,2]. We generated a high-quality dataset for intrusion detection from the Locked Shields live fire cyber defense exercise. This exercise provides a realistic infrastructure with a variety of devices and sophisticated attacks carried out by a “red team”. This ensures our dataset includes a wide range of malicious events and provides the

ground truth. Furthermore, since we document all attacker actions, we can accurately label our data [3–5]. The LSPR23 dataset, originating from the Locked Shields 2023 cyber-defense exercise, contains current attack types, maintains a balance between benign and malicious data, uses state-of-the-art systems, covers attack campaigns executed over an extended period, and exhibits high labeling accuracy, thus addressing the challenges associated with generating a cybersecurity dataset. Data protection is ensured by implementing a Virtual Blue Team (VBT) that primarily observes the network to obtain the dataset, ensuring that no sensitive information is compromised. The recorded network traffic comprises 90% benign events and 10% malicious events, reflecting a realistic representation of network traffic and allowing users the

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<https://doi.org/10.1016/j.jisa.2024.103847>

flexibility to subsample to adjust this distribution, potentially helping to prevent overfitting and improve the generalization capabilities of machine learning models.

Research questions. This work addresses the following research questions:

1. How can a high-quality intrusion detection dataset from a live-fire exercise like Locked Shields be generated and evaluated?
2. How can advanced labeling techniques improve the accuracy of network flow classification in the context of intrusion detection?

These issues are at the heart of contemporary research, underlining the importance of high-quality datasets such as the LSPR23, which stems from the need for accurately labeled data representative of real attack scenarios. Creating the LSPR23 dataset within Locked Shields illustrates an innovative approach to generating intrusion detection training data. The methodology adopted for accurately labeling and balancing benign and malicious data improves the quality of the dataset and answers the question of how high-quality datasets can be generated and evaluated in realistic contexts. Furthermore, the advanced labeling technique used in LSPR23 opens up new possibilities for improving the accuracy of network flow classification, directly addressing the second research question. The composition of the dataset, which includes sophisticated attacks and a variety of network behaviors, provides a solid basis for the development and testing of AI-based intrusion detection systems, highlighting the importance of these data in promoting significant advances in the field of cyber defense.

Challenges. Creating a high-quality machine learning dataset is challenging. In the following, we list the main challenges.

- C1 *Realistic infrastructure:* It requires an extensive and realistic infrastructure that reflects the modeled environment. This involves deploying various types of hardware and software, managing a complex network with different types of traffic, and maintaining a baseline level of user activity. Building and managing such large and complex infrastructure requires significant financial resources, qualified personnel, and advanced technical skills [6, 7].
- C2 *Sophisticated attacks:* The attacks need to go beyond script-generated cyber attacks or incorporate some existing sample of malicious activity. Replicating tactics, techniques, and procedures (TTPs) used by regular as well as sophisticated attackers requires involving cybersecurity specialists and experienced red teams [8, 9].
- C3 *Ground truth:* To accurately label data and understand the motivations and techniques of the attackers, the red team itself has to document this information in great detail. However, such details are often difficult to obtain because attackers may be reluctant to share their strategies and because it can be challenging for a large team to provide information consistently and reliably [10,11].
- C4 *Data protection:* The collected data must be handled with care to ensure that it does not reveal sensitive information that could compromise the privacy or security of the participants. This involves implementing careful data collection and manipulation methods and may even require anonymization or pseudonymization procedures [12,13]. Furthermore, addressing this challenge is also an indicator of the public availability of a dataset for all related works shown in Table 1.
- C5 *Balance:* The dataset must include a well-balanced sample of benign and malicious events. This is important to prevent machine learning models from overfitting to malicious events and to improve their ability to generalize new data. However, generating or collecting a sufficient sample of benign events can be challenging, as it requires simulating or recording a wide range of regular everyday activities [14,15].

- C6 *Large scale:* Large-scale datasets, crucial for machine learning and training, offer a wealth of diverse, real-world scenarios essential for model generalization. They enable models to learn complex patterns and enhance performance. However, their large-scale nature brings significant challenges, including managing high data volumes, real-time processing, scalability of methods, secure storage, robust security measures, privacy preservation, and complex maintenance tasks [16–20].

2. Related work

Sharafaldin et al. [21] highlight the need for up-to-date and realistic IDS datasets for accurate security modeling and threat detection. Inspired by Locked Shields, they proposed a new method for generating such datasets, providing a better understanding of intrusion traffic. Their approach involves creating an IDS dataset based on real-world network traffic, incorporating unique features inspired by the Locked Shields competition. By emphasizing the need for comprehensive and up-to-date datasets that capture the evolving network intrusion landscape, their approach proves highly relevant in the face of rapidly growing cyber threats.

Halisdemir et al. [22] highlight the challenge of obtaining high quality datasets to develop an AI-based IDS. They discuss the limitations of existing IDS datasets available on the Internet and suggested using the unique infrastructure of Locked Shields to generate high-quality datasets. The authors propose integrating a VBT with autonomous agents as a solution to address the security and privacy concerns of blue teams. They also suggested utilizing network traffic directed towards the VBT, which operates independently of any specific blue team, as an effective resource for producing IDS datasets. These recommendations could offer valuable insights into enhancing the quality of datasets in IDS development.

Degirmenci et al. [23] propose a network IDS dataset for a robot operating system. The dataset includes 83 features extracted by CICFlowMeter [1] using the collected PCAP files captured from the traffic of the generated robot system. The authors compare their dataset with the features of six commonly used IDS datasets.

Gehri et al. [24] analyze the underlying cause of the failure of models to generalize under different network conditions. In order to make the analysis, they utilized five distinct network traffic data collected from different teams participating in Locked Shields exercise across different years. The authors claim that the utilization of training data obtained from different environments is significantly more successful compared to training the model using a single dataset. However, the authors could not make their datasets available to the public due to confidentiality concerns.

Another good example of data generation utilizing exercise infrastructure is provided by Tovarnak et al. [25,26]. The authors emphasize that they obtained network traffic flows and event logs by utilizing the infrastructure of the Cyber Czech exercise conducted over a two-day period in March 2019. In this study, raw data is presented. Therefore, researchers are required to process and label the data to create a dataset for model training and testing.

Sangster et al. [27], emphasize that unlabeled network traffic datasets are widely available in the literature, but there is a significant challenge in finding labeled datasets. As a solution to this problem, it is suggested that cyber warfare competitions could be structured to produce modern labeled datasets. They emphasize the importance of the datasets generated by the Defense Advanced Research Projects Agency (DARPA). However, they also articulated the shortcomings of these datasets. They propose a method to generate a labeled dataset from a cyber defense exercise (CDX). The authors emphasize the importance and difficulties of labeling the data. They claim that network warfare games can be used effectively to generate labeled datasets, but the quality of the results depend on the network topology and the sensors used for data capture and labeling.

Table 1

Comparison of related works based on challenges C1–C6. In this table, “Yes” indicates that the study addresses the challenge, “No” indicates that it does not.

Approach	C1	C2	C3	C4	C5	C6	Dataset
TBLGA-99 [28]	No	No	Yes	Yes	No	No	KDDCUP-99
SSTG-12 [29]	Yes	Yes	Yes	Yes	Yes	No	ISCX-12
GGSZ-14 [30]	Yes	No	Yes	Yes	No	No	CTU-13
G-15 [31]	Yes	Yes	Yes	No	No	No	MCFP-15
DMA-16 [32]	Yes	Yes	No	No	No	No	LS-15
SLG-18 [21]	Yes	Yes	Yes	Yes	Yes	No	CICIDS-17
KBHMH-18 [33]	Yes	Yes	No	No	Yes	Yes	LS-17
PKAH-19 [26]	Yes	Yes	No	No	Yes	Yes	RGCE-18
KMST-19 [34]	Yes	No	Yes	Yes	No	Yes	Bot-iot
KALYPK-19 [35]	Yes	No	Yes	Yes	Yes	No	IoT-NID
T5V-20 [25]	Yes	No	No	Yes	No	No	Cyber Czech-19
D-20 [36]	Yes	Yes	Yes	No	No	No	MTAN-20
HTABB-20 [37]	Yes	No	Yes	Yes	Yes	No	MQTT-IoT-IDS-20
MCSAHK-20 [38]	Yes	Yes	Yes	Yes	Yes	No	DAPT-20
LKR-20 [39]	Yes	No	Yes	Yes	Yes	No	DIDarknet-20
GMHL-21 [24]	Yes	Yes	No	No	Yes	Yes	LS-17-LS-21
WT-22 [40]	No	Yes	Yes	Yes	Yes	No	ETFD-22
DKÖY-23 [23]	Yes	Yes	Yes	Yes	Yes	No	Rosid-23
GPE-23 [41]	Yes	No	Yes	Yes	Yes	No	IoT-23
LSPR23 (this work)	Yes	Yes	Yes	Yes	Yes	Yes	LSPR-23

As seen from the literature review, the idea of producing a publicly available and labeled network intrusion detection dataset using exercise network traffic is not a new concept. However, confidentiality issues and the challenge of labeling have always been problems. In our study, we address these challenges by adding a virtual blue team to the Locked Shields exercise, and we solve the labeling problem using our unique techniques.

To provide a clear comparison of our work with related work, we have summarized the key aspects of each approach in Table 1. The table includes indicators such as the use of a realistic infrastructure (C1), inclusion of sophisticated attacks (C2), provision of ground truth (C3), data protection measures and public availability of a dataset (C4), balance between benign and malicious events (C5) all encompassed within a large-scale context (C6).

Technical details on how these issues were addressed in our work are provided in the subsequent sections of the article. As Table 1 illustrates, our work uniquely addresses all these challenges and makes the dataset publicly available, distinguishing it from existing work in the field.

3. Data collection and enrichment

In this section, we first provide a background on Locked Shields (Section 3.1). Following this, we describe how we collected various types of data during the partners run of Locked Shields 2023: network traffic (Section 3.2), data about the Locked Shields environment (Section 3.3) and red team logs (Section 3.3). Then, we describe how we combined this data and enriched it with other information (Section 3.4). Fig. 1 shows an overview of the data collection workflow.

3.1. Background on Locked Shields

Locked Shields is an annual live-fire cybersecurity exercise organized by the North Atlantic Treaty Organization (NATO) Cooperative Cyber Defence Centre of Excellence (CCDCOE) since 2010 [4]. Blue teams (the defenders) are the primary training audience of the exercise. The participating blue teams enhance their skills across various interdisciplinary categories, including real-time cyberdefense, situation assessment, incident response, scenario handling, and ensuring the functionality of their computer systems.

In addition to the blue teams, four additional teams play crucial roles in the exercise. The red team is responsible for conducting cyberattacks; the yellow team deals with situation awareness; the green team

is responsible for developing and maintaining the infrastructure; and the white team handles the overall control of the exercise, including scenario insertion, strategy games, legal matters, media challenges and special investigations. While the preparation starts well in advance, the intense, real-time exercise occurs over two days.

According to the scenario, the blue teams act as rapid-reaction units dispatched to support the fictional country of Berylia, which is involved in a protracted conflict with another fictional country, Crimsonia. Typically, the red team impersonates cyber threats from Crimsonia. A team dedicated to user simulation plays the roles of different characters in the game operating on the systems.

The role of the red team in the exercise involves executing a series of progressively intensifying attacks. Their task is to progress through several phases, from initial access to achieving system persistence, privilege escalation, data collection and exfiltration, and destruction. It is important to note that due to the limited duration of the exercise (two days), the frequency and intensity of these attacks are significantly higher than those in real-world incidents. The red team employs various attack techniques, as classified by the MITRE ATT&CK knowledge base [5]. These techniques include exploiting vulnerabilities in public applications, compromising legitimate user accounts, leveraging vulnerabilities for privilege escalation, moving laterally using remote services, collecting and exfiltrating data from target systems, defacing systems, and launching denial-of-service attacks.

3.2. Network traffic collection

The architecture of the Locked Shields environment (see Fig. 2) facilitated remote capture of traffic from multiple network segments. We used an encapsulated remote switched port analyzer (ERSPAN) to mirror the network traffic of all hosts in our gamenet to an Arkime [42] instance. Using this Arkime instance, we captured network traffic and conducted exploratory data analytics.

3.2.1. Processing

The acquired network traffic was encapsulated through Generic Routing Encapsulation (GRE) by the Locked Shields infrastructure. Within this encapsulation, the IP addresses of the exercise were situated one layer deeper. To support additional tools, such as Suricata and CICFlowmeter, which should operate within the context of the exercise, it was imperative to eliminate this encapsulation layer. The extraction of the encapsulation layer served the dual purpose of exposing the inherent context of the exercise and optimizing the functionality of the used network analysis tools.

3.2.2. Feature extraction

For the extraction of flow-based features, we used CICFlowMeter [1]. We modified CICFlowMeter to dissect network packets for 26 application protocols used in Locked Shields, incorporating this as the (indexed) Service feature as part of the Own Extra Features. We implemented the service feature in CICflowmeter, by consulting source code of Arkime and Suricata, for the most used protocols discovered during the exploratory data analysis process. The list of protocols included in the dataset can be found in Appendix B. For all indexed features in the dataset, lookup tables are provided to reverse the indexed features back to their original values before indexing.

We added two features, Int/Ext and L3/L4, which were introduced by Känzig et al. [43] for the Locked Shields environment. We also utilized their methods for machine learning and for evaluating the trained model. Additionally, we included the Zeek connection state feature, also used in a previous study by Klein et al. [33] on Locked Shields data, to enable the ability to reproduce that study. We implemented this connection state feature in CICflowmeter by consulting the Zeek sourcecode.

From the Suricata EVE source, we extracted the signature ID along with its revision. This enables users of our dataset to precisely identify

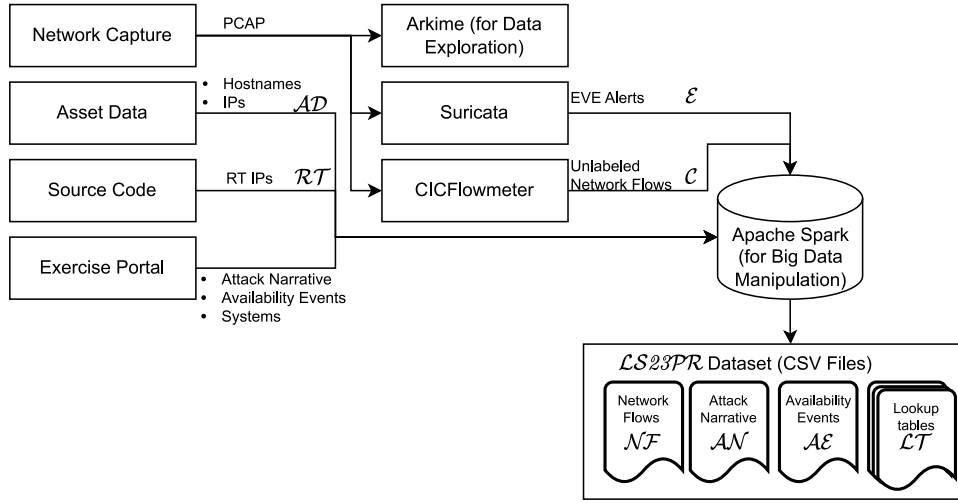


Fig. 1. Data collection flow.

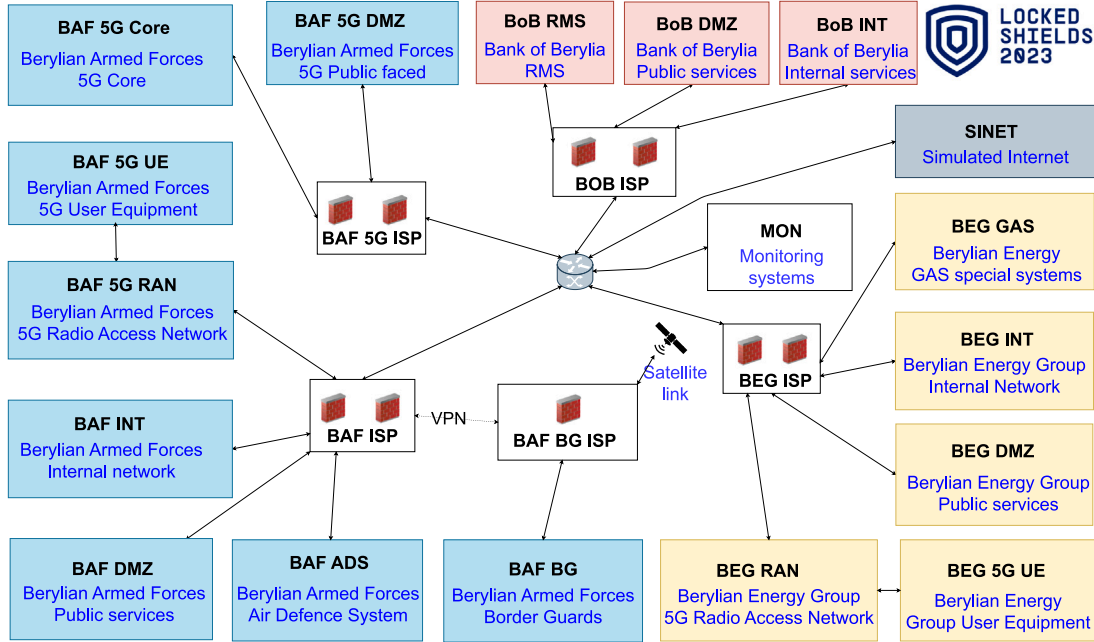


Fig. 2. Locked Shields 2023 network map.

and reproduce the rules we used. In addition, we included the category, severity, and type from the signature-based detection. Suricata also has a separate anomaly detection mechanism that identifies deviations from normal network behavior by analyzing traffic and comparing it to established norms. Moreover, we have incorporated the anomalies detected by Suricata into the features of our dataset. All features in the dataset are indexed and ready to use for machine learning, with the ability to reverse to their original values using the available index tables.

3.3. Environment data

Asset data. The Asset data illustrated in Fig. 1 originates from the vCenter server. For our dataset we systematically polled the vCenter server at regular intervals during the exercise. This polling allowed us to retrieve the dynamic IPs and hostnames of all systems.

Source code. The Locked Shields source code, encompasses all the development resources necessary for creating the exercise environment

used by the organizers. For our dataset, we extracted the complete list of IPv4 and IPv6 addresses that have been duly authorized for use by the red team during the exercise.

Exercise portal collection (EXPO). The Exercise Portal (EXPO) depicted in Fig. 1 serves as the central information hub for the Locked Shields exercise. Accessible via a web page, it aggregates data from multiple sources, including scores, team tickets, various events, and system/service configurations. Attendees utilize EXPO to report their findings, while organizers rely on it to monitor services, check information like system configurations and much more. For our dataset, we extracted attack narratives (Table 2) and availability events (Table 3) from EXPO.

3.4. Data enrichment

In machine learning-based research, enriching a dataset can significantly enhance the effectiveness of the resulting models. In this section, we detail the various processes we employed to enrich our

dataset. We first discuss the enrichment with green team information. Next, we delve into the enrichment with IDS signatures and anomalies, outlining the process of merging network flow features with open-source Suricata IDS rules. This process provides context to the flows and enables research on specific attack types. Finally, we discuss the data labeling process, where each flow in the dataset is labeled, distinguishing between different teams and network segments. These enrichment processes add valuable layers of information to our dataset, enhancing its utility for IDS research. The following subsections provide a detailed exploration of each of these elements.

3.4.1. Enrichment with green team information

We took advantage of the meta information in the Locked Shields exercise that is normally only available to the green team. This information includes the complete internal infrastructure that supports the exercise. More specifically, this includes the mapping of all virtual machines, teams, possible network segments, IP addresses, and domain names that the attacking red teams, the defending blue teams, and the organizing green team could use. While the red team is required to use the preset IP addresses, they are allowed lateral movement from previously compromised hosts to any blue team. Some of the virtual machines used RFC 3041 [44] to generate temporary random IPv6 addresses. We mapped those RFC3041 IPv6 addresses to their host names by means of extracting those specific addresses from the asset data. Finally, using the discovered host names we labeled the flows to the correct team and network segment. During the exercise, the availability of all blue teams' hosts was monitored by the green team. We extracted the availability events from the EXPO and generated features as listed in Table 3. These availability events are particularly useful for users of this dataset as they provide insights into the operational status of the hosts over time. This information can be crucial for understanding the impact of specific attacks on the availability of services and systems. It can also help identify patterns or correlations between certain attacks and changes in availability, providing additional context for the analysis of network flows and attack patterns.

3.4.2. Enrichment with red team information

After the processing of the network traffic with CICFlowmeter, we used the Apache Spark capabilities for manipulating the big data of the network flows. This facilitated systematic labeling of the flows by cross-referencing the IP addresses associated with the red team against those embedded within the network flow. In addition to the source code, we also extracted attack information from the Exercise Portal (EXPO), which imparts a more comprehensive and high-level narrative of the executed attacks compared with the network flow labels. Table 2 presents a detailed presentation of this available information within the dataset and provides a structured overview of the attack-related data from the exercise.

3.4.3. Enrichment with IDS signatures and anomalies

To add more context to the network traffic, we pass it through Suricata [45], an intrusion detection system (IDS). Suricata can passively monitor network traffic and identify potential security threats or attacks.

The IDS information in the LSPR23 dataset shows what is detected without machine learning. This baseline is helpful for analyzing the impact of machine learning.

Suricata requires a set of rules to work. For our analysis, we used the following publicly available rule sets:

- Emerging Threats Open Ruleset [46]: Ruleset designed to detect a broad range of threats, including malware and phishing.
- Threat Hunting Rules [47]: Contains a set of threat-hunting rules.
- Core Malware Rules [48]: Contains a set of core malware rules.
- Stamus Networks Lateral Movement Rules [49]: Contains a set of lateral movement rules that detect potential lateral movement within a network's organization.

Reducing false positive alerts is essential to optimize an intrusion detection system like Suricata. To achieve this, we reviewed the alert logs during a familiarization period where the red team is not active and identified frequently encountered signatures that may generate excessive noise or false positives. Once identified, these signatures can be disabled or filtered to establish a more stable baseline of alerts with less noise. By carefully filtering out frequently encountered signatures that produced false positives, we reduced the noise in our alert logs and focused on alerts that were more likely to indicate genuine security threats. During our experiment, a couple of Suricata signatures were disabled by ID to reduce false positives. We list these rules in Appendix A. While reducing false positives is crucial for minimizing alert fatigue and maintaining operational efficiency, it is equally important to ensure that this does not inadvertently increase the rate of false negatives. An effective intrusion detection system must strike a delicate balance between the two, optimizing its sensitivity to detect real threats without being overwhelmed by false alarms.

3.4.4. Data labeling

Each flow in the dataset contains source and destination labels, which distinguishes between the blue-, green-, red-team, external, and scoring systems. For the blue team, each network segment is also labeled for the source as well as the destination. For example, in [33,43], a flow is considered malicious if the source or destination IP in the flow originates from or connects to the red team infrastructure; all other flows are labeled benign. We also provided the same type of derived "Label" based on source and destination labels. Currently, we cannot label stepping-stone attacks as malicious. Therefore, when the red team compromises a blue-team host and uses it to attack another blue-team host, this is incorrectly labeled as benign. We labeled network flows as malicious if they originated from or were destined to the red team, irrespective of the benign intent of this team. This labeling is based on the perspective of the system and not on human intentions. Furthermore, within the context of the Cyber Kill Chain [50], a visit to a website during the reconnaissance phase, which is typical, is classified as benign. This emphasizes that our flow labeling system is about actions and not about intent. In addition to flow features, the dataset also provides higher-level attack information that does indicate intent. The specific attack type conducted by the red team is available as a separate set of features extracted from the exercise portal (EXPO) and we show them in detail in Table 2.

The network flow features are merged with the open-source Suricata IDS rules using an SQL join query. The Suricata EVE information merged with the flow features can be used for research on specific attack types and provide context to the flows.

Researchers can use the network segment feature to study specific segments of interest. For processing this dataset, we recommend using a big-data framework like Apache Spark with a distributed file system such as Apache Hadoop.

3.4.5. Formal definitions

In this section, we provide formal definitions of enriched and labeled network flows from the LSPR23 dataset. We introduce the symbols that will be used in these definitions. We have a set of IP addresses, IP , involved in the Locked Shields exercise. We also have a set $\mathcal{NP}$ of network protocols (such as TCP, UDP) and a set $\mathcal{PO}$ of machine ports (such as 22, 443). We see time as an integer, with $t_{\min}$ marking the start of Locked Shields Partners Run 23 and $t_{\max}$ marking the end. The interval $[t_{\min}, t_{\max}]$ is represented by $\mathcal{T}$. We begin by defining the LSPR23 dataset, which serves as an overview. Subsequently, we delve into the definitions of the enriched and labeled network flows, which are the primary focus of this section.

Definition 1 (LSPR23 Dataset).

Let us denote the sets included in the LSPR23 Dataset:

$\mathcal{NF}$ = Network Flows

$\mathcal{AN}$ = Attack Narrative

$\mathcal{AE}$ = Availability Events

$\mathcal{LT}$ = Lookup Tables

The *Locked Shields Dataset LSPR23* is a set of these four sets.

$$LSPR23 = \{\mathcal{NF}, \mathcal{AN}, \mathcal{AE}, \mathcal{LT}\}$$

Definition 2 (Unlabeled Flow). An unlabeled network flow c from the set $CICFlowmeter\ C$ is a tuple $(ts_{start}, ts_{end}, IP_{src}, IP_{dest}, Port_{src}, Port_{dest}, IP_{protocol}, Features)$, where

- $ts_{start} \in \mathcal{T}$ is the time at which the flow started;
- $ts_{end} \in \mathcal{T}$ is the time at which the flow ended;
- IP_{src} and IP_{dest} are IP addresses in IP . They are referred to as the ‘source’ and ‘destination’ of the flow, respectively.
- Let $Port_{src}$ and $Port_{dest}$ denote the ports involved in the flow. Specifically, $Port_{src}$ is referred to as the ‘source’ port, while $Port_{dest}$ is referred to as the ‘destination’ port.
- $IP_{protocol}$ (e.g. TCP, UDP) IP Protocol used in this flow
- $Features$ are all other features that are generated by *CICflowmeter*, but not used in further definitions.

Definition 3 (Suricata EVE). Each Suricata EVE alert e of $\mathcal{E}$ is a tuple $(ts, IP_{src}, IP_{dest}, Port_{src}, Port_{dest}, IP_{protocol}, Features)$, where

- $ts \in \mathcal{T}$ is the time at which the alert triggered;
- IP_{src} and IP_{dest} are IP addresses in IP . They are referred to as the ‘source’ and ‘destination’ of the alert, respectively.
- Let $Port_{src}$ and $Port_{dest}$ denote the ports involved in the flow. Specifically, $Port_{src}$ is referred to as the ‘source’ port, while $Port_{dest}$ is referred to as the ‘destination’ port.
- $IP_{protocol}$ (e.g. TCP, UDP) IP Protocol used in this alert
- $Features$ is a set with the Suricata Features (SignatureID_revision, Category, Severity, and Anomaly_event)

Definition 4 (IDS Enriched Flows).

$$IDS = \{(c, e) \in C \times \mathcal{E} : condition(c, e)\}$$

$$condition(c, e) : (IP_{src} \wedge Port_{src} \wedge Time_{condition}(c, e))$$

$$\wedge Protocol_{condition}(c, e) \wedge Time_{condition}(c, e))$$

$$IP_{src} \wedge Port_{src} : \left((IP_{src}(c) = IP_{src}(e) \wedge IP_{dest}(c) = IP_{dest}(e)) \right.$$

$$\wedge (Port_{src}(c) = Port_{src}(e) \wedge Port_{dest}(c) = Port_{dest}(e)) \Big)$$

$$\vee$$

$$\left((IP_{src}(c) = IP_{src}(e) \wedge IP_{dest}(c) = IP_{dest}(e)) \right.$$

$$\wedge (Port_{src}(c) = Port_{src}(e) \wedge Port_{dest}(c) = Port_{dest}(e)) \Big)$$

$$Protocol_{condition} : IP_{protocol}(c) = IP_{protocol}(e)$$

$$Time_{condition} : ts(e) \in [ts_{start}(c), ts_{end}(c)]$$

Definition 5 (Asset Data). Each element ad of $\mathcal{AD}$ is a tuple $(Expoid, IPs, Teamname, Segment)$, where

- $Expoid$ is the id (hostname) of a system in the LS Exercise Portal
- $IPs \subseteq IP$ is the set of IP addresses of a system
- $Teamname \in \{blue, green, red, yellow, white\}$
- $Segment$ is the Blue Team network segment as seen in Fig. 2 if $Teamname = blue$, otherwise $Segment$ is undefined

with the constraint that each $Ip \in IPs$ is associated with one and only one $Expoid$.

Definition 6 (Red Team IPs). Each element rt of $\mathcal{RT} \subseteq IP$ is an external ip address that the Red Team can use and is considered malicious. These ip addresses are mostly not part of the Asset Data.

Definition 7 (Labeled and Augmented Flows $\mathcal{NF}$).

$$\mathcal{NF} = \left\{ \begin{array}{l} (ids, src_Expoid, src_Teamname, src_Segment, src_label, \\ dst_Expoid, dst_Teamname, dst_Segment, dst_label) \\ | ids \in IDS, ad \in AD, \\ \begin{array}{l} src_label = \begin{cases} 0 & \text{if } src_Teamname = \text{"blue"} \text{ or} \\ & src_Teamname = \text{"green"} \\ & \text{and not contains}(src_Expoid, \text{"score"}) \text{ and} \\ & \text{not } IP_{src}(ids) \in \mathcal{RT} \\ 1 & \text{if } IP_{src}(ids) \in \mathcal{RT} \\ 2 & \text{if contains}(src_Expoid, \text{"score"}) \text{ and not} \\ & IP_{src}(ids) \in \mathcal{RT} \\ 3 & \text{otherwise} \end{cases} \\ \\ dst_label = \begin{cases} 0 & \text{if } dst_Teamname = \text{"blue"} \text{ or} \\ & dst_Teamname = \text{"green"} \\ & \text{and not contains}(dst_Expoid, \text{"score"}) \text{ and} \\ & \text{not } IP_{dst}(ids) \in \mathcal{RT} \\ 1 & \text{if } IP_{dst}(ids) \in \mathcal{RT} \\ 2 & \text{if contains}(dst_Expoid, \text{"score"}) \text{ and not} \\ & IP_{dst}(ids) \in \mathcal{RT} \\ 3 & \text{otherwise} \end{cases} \end{array} \right. \\ \text{where } (IP_{src}(ids) \in IPs(ad) \vee IP_{dst}(ids) \in IPs(ad)) \end{array} \right.$$

This definition represents a dataset that enriches *IDS* flows with asset information from $\mathcal{AD}$, while excluding IP addresses post-join. The definition ensures that each flow in *IDS* is augmented with asset data, where source and destination IPs are matched with $\mathcal{AD}$ entries, appending $src_$ or $dst_$ prefixes to the asset columns. Additionally, src_label and dst_label are introduced to indicate the presence of source and destination IPs within the malicious IP set $\mathcal{RT}$. A label of 0 is given to IPs that do not belong to the red team, while a label of 1 indicates malicious activity. Additionally, a label of 2 is assigned when the ip is used for scoring. Conversely, a label of 3 is used for external IPs.

4. Dataset evaluation

In this section, we describe the dataset that resulted from the process described in the previous section. Upon publication of this study, the LSPR23 dataset is publicly available on the Zenodo dataset repository website [51]. This section begins with an overview of the recorded activities of the dataset, following with an overview of the included features. We then present statistics of the dataset. Following the presentation of the statistics, we delve into an analysis of the quality of the dataset, employing both quantitative and qualitative methodologies. Our analysis begins with a robust evaluation of the dataset. To ensure this, we apply the quality criteria outlined by the evaluation framework proposed by Gharib et al. [52] to test the quality of our dataset. Subsequently, we conduct a quantitative examination of the Suricata IDS alerts based on signatures and anomalies. We also assess the reproducibility of machine learning experiments conducted by Känzig et al. [43], who used a private Locked Shields dataset.

4.1. Recorded activities

Fig. 3 provides an overview of the activity during Locked Shields Partners Run 2023. The Partners Run edition of Locked Shields spanned two days. Day 1, known as the familiarization period, involved the blue teams studying the network they were tasked with defending. On day 1, there were no active attacks from the red team, although nondestructive malware beaconing was allowed. Day 2, starting at 09:00 GMT, marked the commencement of active attacks by the red team, as indicated by the peak of malicious activity in Fig. 3.

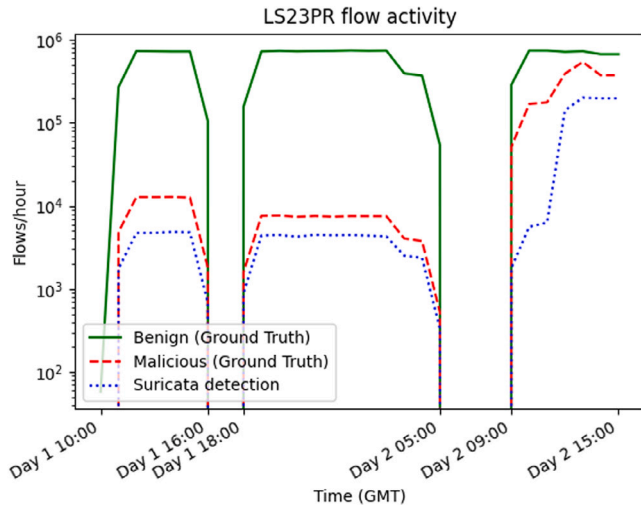


Fig. 3. LS23PR flow activity over time.

The reason for the observed gaps on day 1 (16:00–18:00) and day 2 (05:00–09:00) was technical difficulties with the recording infrastructure. However, as blue and red teams are only allowed to connect to their networks during office hours, the resulting dataset is not missing important information. Starting from 09:00 GMT on Day 2, the red team launched attacks employing various techniques, including:

- Implanted vulnerabilities in network services
- System backdoors
- Network attacks on BGP & ISP
- APT-like command&control using custom malware

4.2. Features

The LS23PR public dataset consists of the following three parts:

- Network flow features (Details in Table B.9).
- Attack narrative features (Details in Table 2).
- Availability event features (Details in Table 3).

4.2.1. Network flow features

The full set of network flow features including its description can be found in Table B.9. Here we present a course overview of the available Network flow features:

- 84 CICFlowMeter features [53]
- 4 Suricata features: SignatureID_revision, Category, Severity, and Anomaly_event
- 2 Features as defined by Känzig et al. [43]: Int/Ext and L3/L4
- 1 Bro/Zeek connection state feature [54]
- 10 Own extra features, providing detected protocol(service), labels and context of the flow

4.2.2. Attack narrative features

The Attack narrative features, including their descriptions, can be found in Table 2.

4.2.3. Availability event features

The Availability event features, including their descriptions, can be found in Table 3.

4.3. Dataset statistics

The captured network trace statistics are available in the Appendix in Table B.10, providing information on the timing and a breakdown of the 445,232,287 network packets and 90,729,702,579 bytes.

Fig. 4 presents the number of network flows for each unique 1466 source and 31,598 destination IP addresses. This demonstrates the effort put into the Locked Shields exercise to include normal external network traffic.

Fig. 5 shows the distribution of the detected services. Here, we notice that the red team primarily uses encrypted Transport Layer Security (TLS) for their activities. Combined with the Hyper Text Transfer Protocol (HTTP), this indicates that web services are favored for malicious activities. In Fig. 6, we present the ports most used by these web services, port 80 for plain text (HTTP) and port 443 (TLS) for encrypted communications. On both of these ports, it can be observed that most of the malicious traffic is directed to the red team. For port 80, it can be noted that the red team is much more present compared to port 443. Combined with the information from Fig. 5, this also suggests that TLS is not only used on port 443.

While encrypted traffic on TCP port 443 is commonplace, we draw attention to Fig. 7, which reveals the existence of malicious traffic on UDP port 443.

Table 4 provides an overview of the available labels and the count of flows in the network segments. It is important to note that the “derived label” in this table is a binary classification, indicating whether a flow is benign or malicious. This label is derived from the source and destination labels, and a flow is marked as “malicious” if either the source or destination label is 1, indicating a connection to the red team infrastructure.

Table 5 presents a top 5 of the performed attacks that are available in the attack narrative features as presented in Table 2.

In Table 6, we observe the number of availability events for the hosts that require protection during the exercise. These events are also utilized to score the blue teams.

4.4. Dataset quality

The quality of a dataset is a critical factor in machine learning-based research, as it directly impacts the reliability and generalizability of the findings. In this section, we detail the various facets that contribute to the quality of our dataset, derived from the Locked Shields 2023 Partners Run infrastructure. Our evaluation of our dataset is based on the criteria outlined by Gharib et al. [52], which provides a comprehensive framework for assessing the quality of intrusion detection datasets. We discuss the comprehensive nature of the network configuration, the completeness of the captured traffic, the meticulous labeling of the dataset, and the breadth of interaction covered. We also address the thoroughness of the capture process, the diversity of available protocols, the variety of attacks, and the heterogeneity of the data. Finally, we outline the extensive feature set computed for each of the recorded network flows. Each of these aspects plays a pivotal role in ensuring the robustness of our dataset and, consequently, the validity of our study. The following subsections provide a detailed exploration of each of these elements, as per the evaluation framework proposed by Gharib et al.

Complete network configuration. The Locked Shields 2023 Partners Run infrastructure is comprehensive, consisting of a mix of information technology (IT) and operational technology (OT). The network topology encompasses various IT systems, such as internet service provider (ISP) technology, a simulated Internet with internal hijacked IP addresses, redundant routers, switches, firewalls, and VPNs. The topology also includes military 5G communication, SCADA/ICS for a power plant, SWIFT banking, military border guard, and satellite communications. In particular, the operating systems used are not limited to Ubuntu and Windows, but encompass a wide range of systems.

Table 2
Attack descriptions in the LSPR23 dataset.

Features	Interpretation
Automatic_verify_time	Timestamp when automatic verified, NULL otherwise
Labels	GOAL, METHOD, and TOOLS used (according to MITRE [5])
Category	Category of the attack WEB, CS, or NET
Subject	Short description of the attack
Description	Full description of the attack
Segments	Array of affected segments
Phase	Phase of the attack
Steps_name	Short name of the attack
Steps_time	Time of the attack
Content	Specific attack-related extra info (related IPs, domain, time, hostname, commands, logs, b64-encoded images)

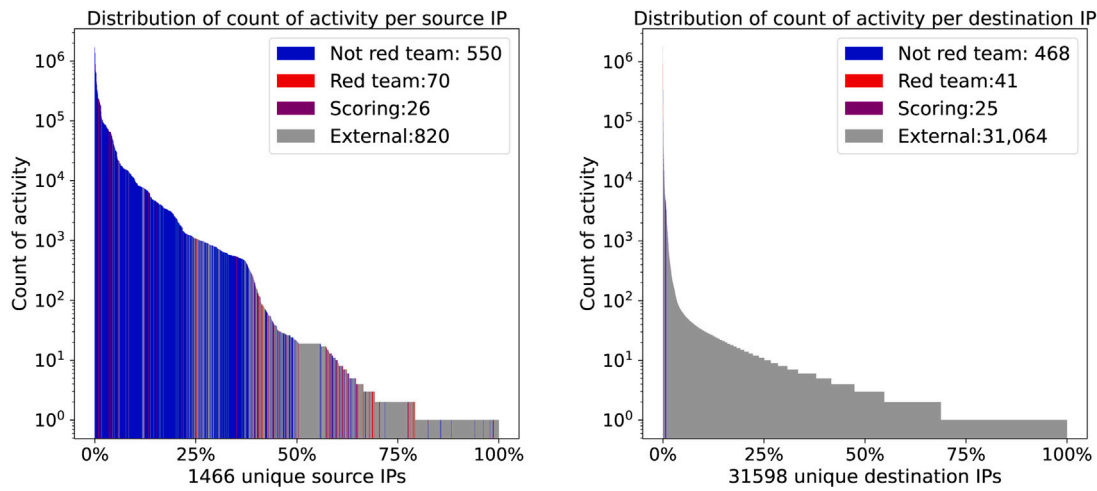


Fig. 4. Distribution of the count of flows per source and destination IP.

Table 3
Availability events.

Features	Interpretation
Change_time	The timestamp of the change of availability
Check_output	Textual output of the availability check
Id	Availability ID
Status	New status (CRITICAL, OK)
Status_old	Previous status (UNKNOWN, CRITICAL, OK)
Expo_id	Exercise portal ID of the host to match with network flow features
Service_name	The service name of the availability check

Complete traffic. The dataset contains traffic from all teams, including the defenders (blue teams), attackers (red teams), organizers (green team, white team, and yellow team), and the user simulation team.

Labeled dataset. In Section 4, we presented the number of benign and attack labels, which serve as the ground truth. Each flow in the dataset is labeled with the ground truth as well as the Suricata signature and anomaly IDs. Currently, we cannot label stepping-stone attacks as malicious. Therefore, when the red team compromises a blue-team host and uses it to attack another blue-team host, this is incorrectly labeled as benign. Without internal malicious labels, it is difficult for machine learning to analyze and test internal flows. Additionally, it is important to note that our dataset has limitations related to the IDS signatures. For instance, signature 2033713, which detects Cobalt Strike Beacon traffic, was mistakenly disabled. This limitation can potentially affect

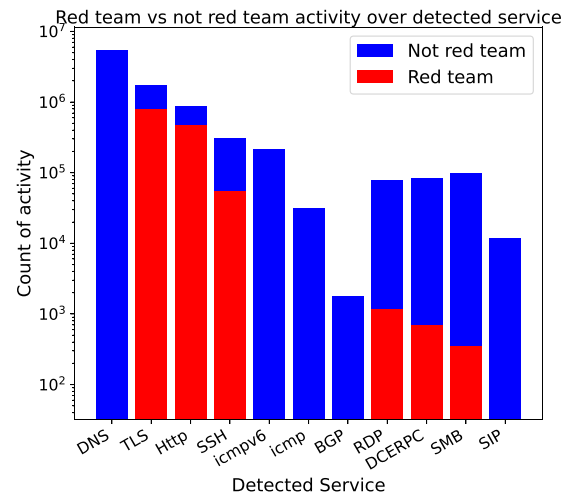


Fig. 5. Distribution of detected services.

the accuracy of the labels and should be taken into consideration when using the dataset.

Complete interaction. The dataset covers all the systems shown in Fig. 2, including systems used for scoring by the green team. It encompasses

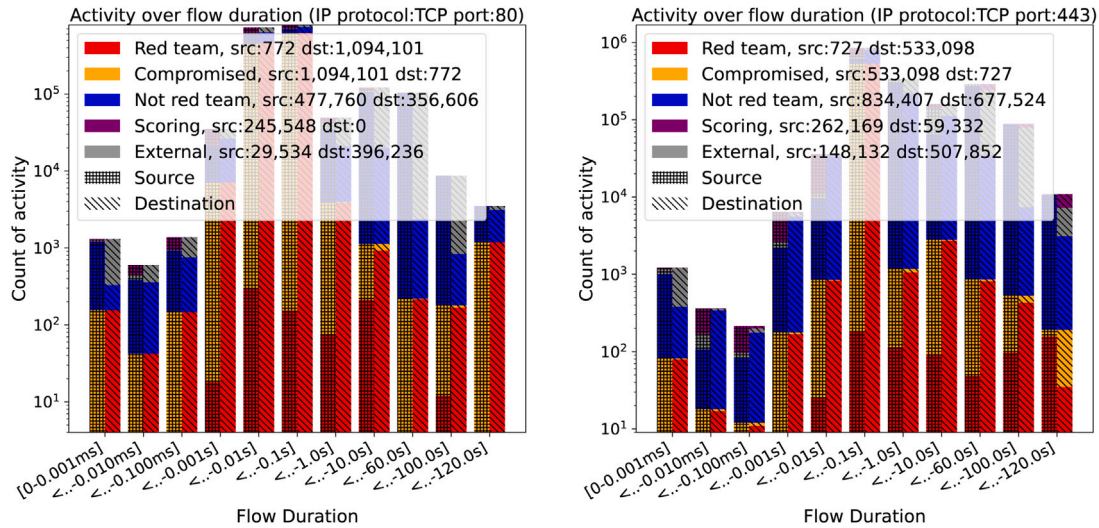


Fig. 6. Flow duration of web services on TCP port 80 and 443.

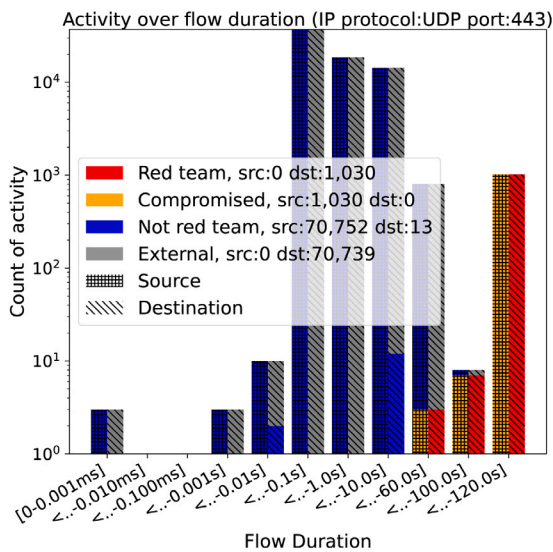


Fig. 7. Distribution of network flows on UDP port 443.

communication between internal LANs and internal ISP communication. For example, this enabled the capture of border gateway protocol (BGP) attacks on the ISP.

Complete capture. All network traffic from all systems was recorded using ERSPAN. After the conclusion of day 1, the blue teams could not interact with their systems, and the red teams were also prohibited from doing so. During this game-closed period, traffic was not captured for about six hours. However, this should not pose a problem for the dataset because many more hours of “idle” data is available during this period. And most importantly, no attacks took place during this period.

Available protocols. In Section 4.5, we provided an overview of the protocols we dissected by modifying CICFlowMeter. In addition to commonly available IT protocols, the dataset includes a broad spectrum of special systems such as 5G, Kubernetes, Satellite, radar, SWIFT, and SCADA/ICS protocols.

Table 4

Label and network segments overview of the LSPR23 dataset.

Total flows	16,353,511	
Labels:	Source	Destination
Blue flows	12,493,826	10,821,533
Red flows	13,880	1,630,719
Green flows (not scoring)	17,256	263,318
Green flows (scoring bots)	3,415,280	136,321
External flows	413,269	3,501,620
Benign flows ^a	14,708,912	
Malicious flows ^a	1,644,599	
Network segments:		
Berylia Energy Group	7,566,863	6,755,988
Berylia Airforce	4,573,708	3,849,478
Bank Of Berylia	277,705	188,984
Berylia Airforce 5G	68,369	27,649
Other	3,866,866	5,531,412

^a Derived labels.

Table 5

Top 5 available attacks in high-level narrative Table 2.

Attack labels “Goals”	
Privilege escalation	63
System compromise	58
Data theft	52
Website defacement	47
Non destructive	18
Attack labels “Methods”	
Remote code execution	35
Authorize with default credentials	28
Remote desktop	23
Authorize with RT credentials	10
Authorize with stolen credentials	9

Table 6

Summary of system availability counts in Table 3.

Availability events	
Critical	12 063
OK	11 729

Attack diversity. The red team consists of subteams focusing on client networks, the web, and special systems. All current technologies are

Table 7

IDS performance compared to ground truth.

Approach	True positive	False positive	Precision	Recall	F1 score
IDS signatures	625,813	88,197	0.876	0.381	0.531
IDS anomalies	28,257	47,789	0.371	0.0174	0.033

targeted. The red team aims to avoid disrupting the systems of the blue teams so that they can continue their attacks. Once the red team achieves their initial objectives, they move to more disruptive attacks. The red team can also instruct the user simulation team to click on a URL and execute malware.

Heterogeneity. We captured network traffic using ERSPAN from all machines. On the low-level we labeled all flows for both the source and destination and enriched these flows using Suricata IDS logs. And on the high-level we provide an attack narrative for longer time periods using MITRE ATT&CK classifications [5].

Feature set. We recorded more than 16 million network flows, computed and enriched 94 features for each of them. The dataset is provided as a set of CSV files in Tables 2, 3 and B.9 including lookup tables for the indexed features and additional metadata.

Metadata. We captured and stored all available metadata from the supporting system of the exercise. We selected the relevant metadata for inclusion in the LSPR23 dataset. This includes the flow information such as timing, IP addresses, network segment, host and domain names and the services running on the hosts. As well as attack information such as timing, IP addresses, target host names, goals, method and used tools.

4.5. IDS accuracy evaluation

In this section, we analyze how well the Suricata IDS features correlate with the ground truth. This provides an intuition regarding what level of detection performance is possible. Precision and recall scores estimate the false positive rate and completeness of the IDS alerts compared to the ground truth labels as described in Section 3.4.4. These metrics help blue teams understand the reliability and effectiveness of Suricata in correctly identifying and classifying network flows. When examining the outcomes in Table 7, it is evident that the Suricata signatures exhibit a seemingly remarkable maximum precision of approximately 0.876, indicating their effectiveness in correctly identifying positive cases. However, their recall rate of 0.381 suggests a limitation in capturing all positive cases, resulting in missed detections. On the other hand, the Suricata anomalies demonstrate a much lower precision value of 0.371 and a recall of 0.017, indicating low detection performance. Because the anomalies scored lower than the signatures, we are only showing the detection results using signatures in Fig. 3.

4.6. Reproducibility evaluation

Reproducibility is a cornerstone of scientific research. It validates the findings of a study by independently repeating the experiments and comparing the results. In this context, we aimed to reproduce the work of Känzig et al. who used a private dataset compiled from the Locked Shields exercise. Our study stands out as we provide a publicly available dataset, enabling other researchers to reproduce and validate previous studies that used Locked Shields data. The task at hand was essentially a binary classification task, using the “derived label” from Table 4. To adhere closely to the methodology of Känzig et al. we conducted machine learning using a random forest model. We selected the same model parameters and features as Känzig et al. ensuring a fair comparison. This includes using the same train–test split ratio of 4:1 for our dataset, ensuring a fair and direct comparison of results. Referring to the scores in Table 8, our approach achieved 326,836 positives with

Table 8

Reproduction analysis using ML/random forest.

Approach	True positive	False positive	Precision	Recall	F1 score
This dataset	326,836	69	0.999	0.994	0.997
Känzig et al. [43]	–	–	0.99	0.98	0.984

only 69 false positives. This resulted in a precision of 0.999, a recall of 0.994, and an F1 score of 0.997. These scores closely approach those reported by Känzig et al. who achieved a precision of 0.99, a recall of 0.98, and an F1 score of 0.984. These results demonstrate that it is indeed possible to reproduce the work of Känzig et al. using our dataset. Moreover, our approach not only reproduces their work but also provides comparable results, which could be attributed to our more accurate labeling methods and the quality of our dataset.

5. Discussion

This paper presents a labeled dataset, an important step towards the vision of an automated defender in cyber-defense exercises. However, there is still a long road ahead to take to achieve the ultimate goal of an intelligent and autonomous cybersecurity agent as introduced in [55]. In this section, we discuss the missing pieces and how our dataset helps to address them.

We identify four main areas that require additional research:

- *Data collection and analysis:* While our dataset focuses on network traffic, there are many more types of data that can and should be collected and transformed into a new dataset. Examples include log information from hosts, alerts from security appliances, and user interactions.
- *Intelligence:* With the existence of an initial dataset, it is now possible to develop AI models that can learn from this data and apply their intelligence to defend other networks.
- *Actions:* In addition to developing intelligence that determines what actions to take, there is also a need for further work to implement and execute these actions.
- *Interfaces:* In a cyber defense exercise like Locked Shields, communication with other teams is part of the tasks performed by a blue team. However, in the context of an autonomous agent, this task is more challenging because autonomous agents do not inherently possess the ability to communicate with humans.

Below, we provide more details about the remaining work.

Data collection and analysis. While most malicious activities induce network traffic, detecting them based on this traffic is often challenging, especially when the traffic is encrypted and the communication contents cannot be easily known.

To improve detection accuracy, researchers should have additional data available. This could include logging data from all hosts in the gamenet. Another step is to capture more detailed logging from the red teams with the aim of correlating their activities with the internal flows to correctly label stepping-stone attacks.

We also see potential for future work based on the network traffic dataset that we publish here. Our preliminary analysis is limited to detecting red team traffic using the same features as in [43]. A more detailed analysis could investigate the specific actions of the red team in more detail, potentially in conjunction with the Suricata signatures and anomalies included in the dataset. Although we lack stepping-stone labels in this dataset, we do see potential for future research on carefully analyzing the internal flows in the time window of the labeled attacks from the compromised hosts, which could produce stepping-stone labels. In Locked Shields, both attackers and blue team systems can have multiple IP addresses, complicating network traffic analysis. The use of the hostname feature can facilitate this analysis

by providing a more detailed and focused view of the internal flows, thereby improving the identification of compromised hosts.

It would also be beneficial if other teams in Locked Shields or similar exercises could collect data.

Intelligence. The greatest potential for future work lies in developing more sophisticated models that can provide the intelligence needed to detect, prevent, and recover from attacks. Possible future research directions include:

- Supervised machine learning techniques to detect potentially malicious activities, such as leveraging network traffic, as demonstrated in Section 4.5. However, there is room for improvement, such as incorporating more features and selecting features that are less dependent on the specific environment.
- Unsupervised machine learning techniques to detect anomalies across all layers and abstractions, including network traffic and host logs.

Actions. In addition to models and intelligence that determine the actions of the virtual blue team, there is a need for components that can effectively perform these actions. Future work in this area primarily involves engineering efforts to implement various actions, restoring systems, configuring permissions, blocking traffic, killing services, or changing configurations.

Interfaces.

- Large Language Models (LLMs): LLMs [56], like ChatGPT, have recently gained popularity and shown impressive results. These models can be applied directly to communicate with the user simulation team, but they also have the potential to analyze network traffic or system logs. However, further research is needed in this direction.

6. Conclusion

In this study, we present a publicly accessible IDS dataset, derived from the network infrastructure of the Locked Shields exercise. Locked Shields is globally recognized as the largest live-fire cybersecurity exercise. To solve confidentiality challenges, we implemented a virtual blue team during the exercise, resulting in the creation of the LSPR23 dataset from this team's network traffic. This development enables researchers interested in the Locked Shields exercise – or cybersecurity in general – to conduct reproducible research, a feat previously unattainable. By offering our dataset to the public, we aspire to foster a culture of transparency and reproducibility within the field. We extend an invitation to fellow researchers to employ our dataset for the validation and expansion of existing work, thereby advancing the field of machine learning-based detection of malicious traffic. The unique infrastructure of the Locked Shields exercise, the utilization of cutting-edge network devices and hardware components, and the inclusion of the latest attack types executed by the red team, all contribute to the value of the generated dataset. We anticipate that this dataset will significantly bolster IDS research and development efforts in both academia and industry. Among others, it provides a basis for working towards an automated cyber defender – a topic that is of wide interest [55,57,58] and previously not achievable because of the lack of high-quality training data.

CRedit authorship contribution statement

Allard Dijk: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Investigation, Data curation, Conceptualization. **Emre Halisdemir:** Writing – review & editing, Writing – original draft. **Cosimo Melella:** Writing – review & editing, Writing – original draft. **Alari Schu:** Writing – original draft, Investigation. **Mauno Pihelgas:**

Writing – review & editing, Writing – original draft, Validation, Resources, Data curation, Conceptualization. **Roland Meier:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Will be made available on [10.5281/zenodo.8042347](https://doi.org/10.5281/zenodo.8042347).

Appendix A. Ignored IDS rules

- 2200073, 2200074, 2200075, 2200077, and 2200078: These signatures detect invalid checksums for various transport and IP protocols. While they can help detect specific attacks, they can also produce false positives due to legitimate traffic with invalid checksums. Disabling these signatures can reduce the number of false positive alerts related to checksum errors.
- 2210044: This signature detects packets with invalid timestamps, which can indicate potential attacks. However, it can also produce false positives due to clock sync issues or other factors. Disabling this signature can help reduce false positives related to invalid timestamps.
- 2210010: This signature detects three-way handshake problems with incorrect sequence or acknowledgment numbers, which may indicate TCP session hijacking attacks. However, it can also produce false positives due to legitimate network behavior. Disabling this signature can help reduce false positives related to handshake issues.
- 2033713: This signature detects Cobalt Strike Beacon traffic, a common indicator of malware activity. However, it can also produce false positives due to legitimate use of the Cobalt Strike tool. Disabling this signature can help reduce false positives related to Cobalt Strike traffic. We decided to disable this signature because the rule fired many times during testing before Day 1 and was believed to be a false positive alert, but in hindsight, this was an incorrect decision. The red team was also testing their Cobalt Strike setup, resulting in the firing of this rule.
- 2018358: This signature detects suspicious POST requests to a quadruple IP address pointed using a fake browser user agent. While it can assist in detecting specific attacks, it can also produce false positives due to legitimate web traffic. Disabling this signature can help reduce false positives related to suspicious POST requests.
- 2260001: This signature detects the first data in the wrong direction for an application layer protocol. While it can assist in detecting specific attacks, it can also produce false positives due to legitimate network behavior. Disabling this signature can help reduce false positives related to application-level issues.

Appendix B. Features

See [Tables B.9](#) and [B.10](#).

Table B.9

Features in the LSPR23 dataset.

Features	Interpretation
CICflowmeter [53]	
Flow ID	String concatenation of SrcIP, DstIP, SrcPort, DstPort and Protocol
SrcIP	Source IP (exploded format for IPv6)
DstIP	Destination IP (exploded format for IPv6)
SrcPort	TCP/UDP Source Port
DstPort	TCP/UDP Destination Port
Protocol	IP Protocol number
Flow Duration	Flow duration in microseconds
Flow Bytes/s	Number of flow bytes per second
Flow Packets/s	Number of flow packets per second
Tot Fwd Pkts	Total packets in Forward direction
Tot Bwd Pkts	Total packets in the backward direction
Active Mean	Mean time a flow was active before becoming idle
Total Length of Fwd Packet	Total size of packet in forward direction
Total Length of Bwd Packet	Total size of packet in backward direction
Fwd Packet Length Min	Minimum size of packet in forward direction
Fwd Packet Length Max	Maximum size of packet in forward direction
Fwd Packet Length Mean	Mean size of packet in forward direction
Fwd Packet Length Std	Standard deviation size of packet in forward direction
Bwd Packet Length Min	Minimum size of packet in backward direction
Bwd Packet Length Max	Maximum size of packet in backward direction
Bwd Packet Length Mean	Mean size of packet in backward direction
Bwd Packet Length Std	Standard deviation size of packet in backward direction
Flow IAT Mean	Mean packet inter-arrival time in flow
Flow IAT Min	Min time between two packets sent in the flow
Flow IAT Max	Max time between two packets sent in the flow
Flow IAT Stddev	Standard deviation time between two packets in the flow
Fwd IAT Min	Min packet inter-arrival time between two packets in forward direction
Fwd IAT Max	Max packet inter-arrival time between two packets in forward direction
Fwd IAT Mean	Mean packet inter-arrival time between two packets in forward direction
Fwd IAT Std	Standard deviation time between two packets in forward direction
Fwd IAT Tot	Total time between two packets sent in the forward direction
Bwd IAT Min	Min packet inter-arrival time between two packets in backward direction
Bwd IAT Max	Max packet inter-arrival time between two packets in backward direction
Bwd IAT Mean	Mean packet inter-arrival time between two packets in backward direction
Bwd IAT Std	Standard deviation time between two packets in backward direction
Bwd IAT Tot	Total time between two packets sent in the backward direction
Fwd PSH flags	Number of times the PSH flag was set in packets traveling in the forward direction (0 for UDP)
Bwd PSH Flags	Number of times the PSH flag was set in packets traveling in the backward direction (0 for UDP)
Fwd URG Flags	Number of times the URG flag was set in packets traveling in the forward direction (0 for UDP)
Bwd URG Flags	Number of times the URG flag was set in packets traveling in the backward direction (0 for UDP)
Fwd Header Length	Total bytes used for headers in the forward direction
Bwd Header Length	Total bytes used for headers in the backward direction
FWD Packets/s	Number of forward packets per second
Bwd Packets/s	Number of backward packets per second
Packet Length Min	Minimum length of a packet
Packet Length Max	Maximum length of a packet
Packet Length Mean	Mean length of a packet
Packet Length Std	Standard deviation length of a packet
Packet Length Variance	Variance length of a packet
FIN Flag Cnt	Flag count FIN in both directions
SYN Flag Cnt	Flag count SYN in both directions
RST Flag Cnt	Flag count RST in both directions
PSH Flag Cnt	Flag count PSH in both directions
ACK Flag Cnt	Flag count ACK in both directions
URG Flag Cnt	Flag count URG in both directions
CWR Flag Cnt	Flag count CWR in both directions
ECE Flag Cnt	Flag count ECE in both directions
Down/Up Ratio	Download and upload ratio
Average Packet Size	Average size of packet
Fwd Segment Size Avg	Average size observed in the forward direction
Bwd Segment Size Avg	Average size observed in the backward direction
Fwd Bytes/Bulk Avg	Average number of bytes bulk rate in the forward direction
Fwd Packet/Bulk Avg	Average number of packets bulk rate in the forward direction
Fwd Bulk Rate Avg	Average number of bulk rate in the forward direction
Bwd Bytes/Bulk Avg	Average number of bytes bulk rate in the backward direction
Bwd Packet/Bulk Avg	Average number of packets bulk rate in the backward direction
Bwd Bulk Rate Avg	Average number of bulk rate in the backward direction
Subflow Fwd Packets	The average number of packets in a sub flow in the forward direction
Subflow Fwd Bytes	The average number of bytes in a sub flow in the forward direction
Subflow Bwd Packets	The average number of packets in a sub flow in the backward direction
Subflow Bwd Bytes	The average number of bytes in a sub flow in the backward direction

(continued on next page)

Table B.9 (continued).

Features	Interpretation
Fwd Init Win bytes	The total number of bytes sent in initial window in the forward direction
Bwd Init Win bytes	The total number of bytes sent in initial window in the backward direction
Fwd Act Data Pkts	Count of packets with at least 1 byte of TCP data payload in the forward direction
Fwd Seg Size Min	Minimum segment size observed in the forward direction
Active Min	Minimum time a flow was active before becoming idle
Active Mean	Mean time a flow was active before becoming idle
Active Max	Maximum time a flow was active before becoming idle
Active Std	Standard deviation time a flow was active before becoming idle
Idle Min	Minimum time a flow was idle before becoming active
Idle Mean	Mean time a flow was idle before becoming active
Idle Max	Maximum time a flow was idle before becoming active
Idle Std	Standard deviation time a flow was idle before becoming active
Features from Suricata	
SigID_revision	Combination of the Suricata signature ID and used revision
Category	The category of the triggered Suricata rule
Severity	The severity level of the triggered Suricata rule
Anomaly_event	Unexpected conditions such as truncated packets, packets with invalid values, events that render the packet invalid for further processing or unexpected behaviors.
Features from Känzig et al. [43]	
L3/L4 Protocol	0 for TCP, 1 for UDP, 2 for ICMP
Int/Ext Dst IP	0 if Dest IP of a flow belongs to Blue teams subnets, 1 if external
Features from Bro/Zeek [54]	
Conn_state	TCP connection state: S0, Connection attempt seen, no reply; S1, Connection established, not terminated (0 byte counts); SF, Normal establish & termination (>0 byte counts); REJ, Connection attempt rejected; S2, Established, ORIG attempts close, no reply from RESP; S3, Established, RESP attempts close, no reply from ORIG; RSTO, Established, ORIG aborted (RST); RSTR, Established, RESP aborted (RST); RSTOS0, ORIG sent SYN then RST; no RESP SYN-ACK; RSTRH, RESP sent SYN-ACK then RST; no ORIG SYN; SH, ORIG sent SYN then FIN; no RESP SYN-ACK ("half-open"); SHR, RESP sent SYN-ACK then FIN; no ORIG SYN; OTH, No SYN, not closed. Midstream traffic. Partial connection
Own extra features	
Service	One of the following detected protocols: UNKNOWN, DNS, TLS, HTTP, SSH, ICMPv6, SMB, DCERPC, RDP, ICMP, SIP, RTP, SCTP, BGP, RTCP-RRRTCP-SR, RTCP-RR, DHCPv6, SMTP, NTP, LDAP, KRB5, SIP, RTCP-SR, RTCP-APP,RTCP-SDES, RTCP-SRRTCP-RR, or MYSQL
Label_src	(source) 0 if not red team, 1 if red team, 2 scoring, 3 external
Label_dst	(destination) 0 if not red team, 1 if red team, 2 scoring, 3 external
Label	0 if Benign and 1 if Malicious. Derived from the red team condition in Label_src and Label_dst
External_src	1 if source ip is external to the blue team, 0 if not
External_dst	1 if destination ip is external to the blue team, 0 if not
Segment_src	The segment of the BT-network-source in the LS environment. baf_5g_core,baf_bg, baf_ran, bafbg_isp, baf_ads,baf_int, baf_isp, baf_dmz, 5g_isp, 5g_dmz, bob_isp, bob_rms, bob_int, bob_dmz, beg_isp, beg_gas, beg_dmz, beg_ran, beg_int, sinet, mon, None
Segment_dst	The segment of the BT-network-destination in the LS environment.
Expoid_src	Unique identifier for BT-network source host in the LS Exercise Portal(Expo)
Expoid_dst	Unique identifier for BT-network destination host in the LS Exercise Portal(Expo)

Table B.10

Captured network trace statistics.

Statistic	Value
Min timestamp	1,678,274,121,011,931
Trace in seconds	102,394.695
Trace in minutes	1706.578
Trace in hours	28.443
Trace in days	1.185
Gap (Exercise break) without traffic (s)	21,600
Net trace time in seconds (minus gap time)	80,794.695
Start time	2023-03-08 11:15:21.011931
End time	2023-03-09 15:41:55.707294
Number of flows	16,353,511
Ippackets	445,232,287
Ipv4packets	339,039,275
Ipv6packets	106,193,012
Ibbytes	90,729,702,579
Ipv4bytes	69,173,457,796
Ipv6bytes	21,556,244,783
Accumulated flowDuration (s)	120,893,257
Accumulated flowDurationipv4 (s)	82,669,432
Accumulated flowDurationipv6 (s)	38,223,825
Flows per second	202.361
Bytes per second	1,122,966.083
Bytes per second ipv4	856,163.359
Bytes per second ipv6	266,802.724

Table B.10 (continued).

Statistic	Value
Megabits per second	8.984
Megabits per second ipv4	6.849
Megabits per second ipv6	2.134
Packets per second	5510.662
Packets per second ipv4	4196.306
Packets per second ipv6	1314.356

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